Why Does $\text{proj}_{u_1} v + \text{proj}_{u_2} v$ Require $u_1 \cdot u_2 = 0$?

NEU 314

Recall that we defined the projection of v onto u as

$$\text{proj}_u v = \frac{v \cdot u}{u \cdot u} u.$$

When $\{u_1, u_2\}$ is an *orthogonal* basis for a subspace, we can recover v exactly by summing its projections:

$$v = \text{proj}_{u_1} v + \text{proj}_{u_2} v.$$

A natural question is: what goes wrong if $u_1 \cdot u_2 \neq 0$?

The Algebraic Reason

Suppose we want to write $v = c_1 w_1 + c_2 w_2$ for some vectors w_1, w_2 and unknown scalars c_1, c_2. To find c_1, dot both sides with w_1:

$$v \cdot w_1 = c_1(w_1 \cdot w_1) + c_2(w_2 \cdot w_1).$$

- If $w_1 \cdot w_2 = 0$: the second term vanishes, and we get $c_1 = \dfrac{v \cdot w_1}{w_1 \cdot w_1}$ directly – exactly the projection formula.
- If $w_1 \cdot w_2 \neq 0$: the coefficient c_1 depends on c_2 as well. The two are *coupled*, and we cannot read them off independently using the projection formula.

Using the projection formula when $w_1 \cdot w_2 \neq 0$ silently ignores the coupling term, which is the source of the error.

A Concrete Example

Let $w_1 = (1, 0)$ and $w_2 = (1, 1)$, so $w_1 \cdot w_2 = 1 \neq 0$. Take $v = (1, 0)$: this vector points purely in the x-direction. The correct decomposition should assign all of v to w_1 and nothing to w_2.

Computing the projections:

$$\text{proj}_{w_1} v = \frac{v \cdot w_1}{w_1 \cdot w_1} w_1 = \frac{1}{1}(1, 0) = (1, 0),$$

$$\text{proj}_{w_2} v = \frac{v \cdot w_2}{w_2 \cdot w_2} w_2 = \frac{1}{2}(1, 1) = \left(\tfrac{1}{2}, \tfrac{1}{2}\right).$$

Their sum is

$$\text{proj}_{w_1} v + \text{proj}_{w_2} v = \left(\tfrac{3}{2}, \tfrac{1}{2}\right) \neq (1, 0) = v.$$

The sum has acquired a y-component even though v lies entirely on the x-axis! The reason: $w_2 = (1, 1)$ contains an x-component, so projecting v onto w_2 *re-captures* part of the x-direction that was already counted in $\text{proj}_{w_1} v$. The x-direction is counted twice.

Gram–Schmidt Removes the Overlap

Starting from the same pair $\{w_1, w_2\}$, Gram–Schmidt produces an orthogonal pair spanning the same subspace.

Step 1. Keep w_1: set $u_1 = (1, 0)$.

Step 2. Subtract from w_2 its component along u_1:

$$u_2 = w_2 - \text{proj}_{u_1} w_2 = (1, 1) - \frac{(1, 1) \cdot (1, 0)}{(1, 0) \cdot (1, 0)}(1, 0) = (1, 1) - (1, 0) = (0, 1).$$

Now $u_1 \cdot u_2 = 0$. Applying the projection formula to the same $v = (1, 0)$:

$$\text{proj}_{u_1} v + \text{proj}_{u_2} v = \frac{1}{1}(1, 0) + \frac{0}{1}(0, 1) = (1, 0) = v. \quad \checkmark$$

The projection onto $u_2 = (0, 1)$ correctly contributes zero: v has no y-component, and u_2 measures *only* the y-direction. Gram–Schmidt stripped the x-part out of w_2, ensuring that u_1 and u_2 each measure something genuinely independent.

Summary

Orthogonality ($u_1 \cdot u_2 = 0$) guarantees that each basis vector measures a direction *no other basis vector can see*. When this fails, the projection formula double-counts the shared directions, and the sum of projections no longer reconstructs v.